

Popper: Automated Hypothesis Validation with Agentic Sequential Falsifications

How an LLM-agent system turns Karl Popper's "you can only refute, never prove" into a running algorithm — proposing falsification experiments against a free-form hypothesis, executing them as code over real data, and aggregating the evidence with a *sequential statistical test* that holds the false-discovery rate below a knob you set (α), no matter how many experiments it runs or when it decides to stop. Where Robin and AI-Scientist *generate* hypotheses, Popper is the missing other half: the rigorous *validation* engine, and it works in biology, economics, and sociology alike.

Background

Primary paper: Automated Hypothesis Validation with Agentic Sequential Falsifications (Kexin Huang, Ying Jin, Ryan Li, Michael Y. Li, Emmanuel Candès, Jure Leskovec — Stanford CS / Statistics / Data Science Initiative / Harvard, Feb 2025). Code: github.com/snap-stanford/POPPER.

Popper sits at the intersection of two lineages — one philosophical, one statistical — and the whole contribution is making them meet inside an agent loop.

Lineage 1 — Popperian falsification (the philosophy of science).

- **Karl Popper**, *The Logic of Scientific Discovery* (1959/2005). The founding claim: scientific theories can never be *verified* by evidence, only *falsified*. A theory earns credibility by surviving sincere attempts to refute it. Popper's machine implements this literally — it never tries to prove a hypothesis directly; it repeatedly tries to break the hypothesis's logical consequences and reports the hypothesis as "validated" only when those refutation attempts keep failing.
- **Thomas Kuhn**, *The Structure of Scientific Revolutions* (1962); **Imre Lakatos**, *The Methodology of Scientific Research Programmes* (1978). Kuhn stressed the sociotechnical embeddedness of science; Lakatos's "research programmes" evaluate hypotheses inside a web of *auxiliary assumptions*. The paper explicitly maps Lakatos's auxiliary assumptions onto its own "dataset relevance is a prerequisite for a valid test" check (the relevance checker, below). Also cited: van Fraassen's constructive empiricism (empirical adequacy over ontological truth — mirrors Popper's focus on *observable implications* rather than the abstract claim), and Goodman's "grue" paradox on the risk of inductive generalization.
- The reading: Popper-the-system is a 60-year-old epistemology compiled into Python. The novelty is not the philosophy, it is making falsification *automatic, scalable, and statistically honest*.

Lineage 2 — e-values, safe testing, and sequential inference (the statistics). This is the genuinely new machinery and the part to slow down on.

- **Neyman–Pearson (1928, 1933) and Fisher (1936)** — classical Type-I error control. The fixed-sample frequentist baseline Popper must respect.
- **Robbins / Ville** → **martingale tests** — the idea that you can test continuously and stop whenever, if your evidence statistic is a non-negative (super-)martingale.
- **Shafer**, *The language of betting* (arXiv 1903.06991, 2019) and **Grünwald, de Heide & Koolen**, *Safe testing* (arXiv 2020 ITA, 2020) — recast hypothesis testing as betting against the null; the wealth process is an *e-process*. This is where "any-time validity" and "optional stopping is allowed" come from.
- **Vovk & Wang**, *E-values: calibration, combination and applications* (*Annals of Statistics* 49(3), 2021) — the definitional paper for **e-values** and the **p-to-e calibrator** Popper uses verbatim. Read this one if you read only one statistics reference.

- **Wang & Ramdas**, *False discovery rate control with e-values* (JRSS-B 84(3), 2022) — the e-BH procedure; Popper points to it as the route to FDR control over many hypotheses (future work).

Contemporary "AI scientist" systems Popper is positioned against (and complements): *The AI Scientist* (Lu et al., Sakana, 2408.06292) — end-to-end automation in computational ML; *AI Co-Scientist* (Gottweis et al., Google, 2502.18864) — multi-agent hypothesis generation/debate; *Robin* (Ghareeb et al., FutureHouse, 2505.13400) — closed-loop wet-lab biology discovery; *DiscoveryBench* (Majumder et al., 2407.01725) and *TargetVal*-style benchmarks supply the test hypotheses. Every one of these is generation-heavy. Popper is the only one whose deliverable is a *statistically valid verdict* on a hypothesis it did not have to generate.

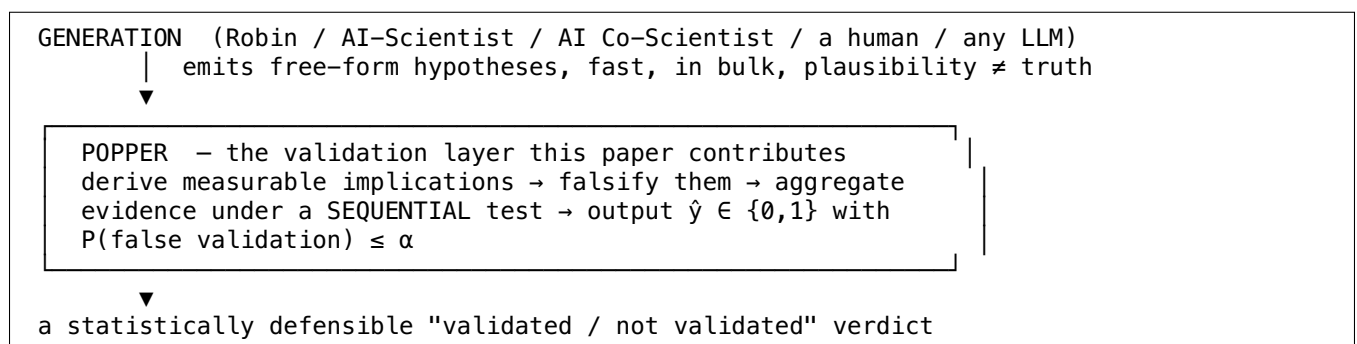
What Problem Does Popper Solve?

The recent explosion of LLM-generated hypotheses created a validation bottleneck. LLMs produce plausible-sounding hypotheses *in volumes that make manual checking impossible*, and their plausibility is uncorrelated with truth (hallucination). The hypotheses that matter are also usually **abstract**: "Gene VAV1 regulates IL-2 production in immune tissue," "a marketing intervention increases retention," "a social policy reduces recidivism." You cannot test those statements *as stated* — there is no single dataset whose column is literally "is this hypothesis true."

So you do what a scientist does: derive **measurable implications** (concrete consequences that must hold *if* the hypothesis is true), test those, and accumulate evidence. Two failure modes lurk:

1. **The generation systems stop short of rigor.** Robin, AI-Scientist, and AI Co-Scientist are brilliant at proposing and even running experiments, but their "this looks supported" judgment is an LLM vibe, not a controlled statistical decision. Nothing stops them from declaring victory on noise.
2. **Naively testing many implications inflates false positives.** Run 20 tests, take the best p-value, and you will "validate" a false hypothesis far more than α of the time. This is the multiple-testing / optional-stopping trap, and an agent that decides *adaptively* when to stop testing makes it worse, not better.

Popper targets exactly this gap: **rigorous, scalable validation of free-form natural-language hypotheses, with a provable cap on the rate of false validations**, while still letting the agent gather evidence adaptively and stop early. The headline framing for a non-bio reader: this is a *domain-general validation layer* you can bolt onto any hypothesis generator.



Key Terms

Hypothesis H (free-form): a natural-language statement defining a relationship r between variables V under context c . Example: $H = \text{"Gene VAV1 regulates IL-2 production in immune tissue"} \rightarrow V = \{\text{VAV1, IL-2 production}\}, r = \text{"regulate"}, c = \text{"immune tissue."}$ Too abstract to test directly.

Null / alternative (main level): H is paired with a **main null H_0** — a whole *family* P_0 of data-generating distributions consistent with "the relationship does not hold" (e.g., "VAV1 does *not* regulate IL-2"). Validating H means *rejecting* H_0 in favor of the alternative. (Same null-vs-alternative logic as any test you know; the twist is H_0 is a family, and the data could be anything an agent can fetch or collect.)

Measurable implication / sub-hypothesis: a concrete, testable consequence that must hold *if* H is true. If VAV1 truly regulates IL-2, then VAV1 should be *preferentially expressed in immune tissue vs. unrelated tissue*. Each implication becomes a **falsification experiment** with its own null sub-hypothesis h_i^0 ("no difference in expression") and alternative h_i^1 . You then run a standard statistical test (t-test, Mann–Whitney U, permutation, χ^2 ...) and get a p-value p_i .

The implication assumption (Assumption 1, the load-bearing one): *if the main null H_0 is true, then every null sub-hypothesis h_i^0 is also true*. This is what lets refuting sub-hypotheses count as evidence against H_0 with honest error control. If you test an implication that does *not* actually follow from H, you can "falsify" it by luck and wrongly validate H — which is precisely why Popper needs a relevance checker.

p-value p_i : as you know — under h_i^0 , $P(p_i \leq t) \leq t$ (super-uniform). The output of one falsification experiment.

e-value e_i : a non-negative random variable with $E[e_i] \leq 1$ **under the null**. Read it as the *payoff of a bet against the null*: a fair bet has expected value ≤ 1 if the null holds, so a large realized e-value (say 30) is strong evidence the null is false. E-values are the currency Popper aggregates because — unlike p-values — they **multiply** and survive **optional stopping**. (Mental model: an e-value is "how many times your money you'd have made betting against H_0 ," and you can only consistently multiply your money if H_0 is really false.)

p-to-e calibrator: a function turning a valid p-value into a valid e-value. Popper uses the Vovk–Wang family $e_i = \kappa \cdot p_i^{(\kappa-1)}$, $\kappa \in (0, 1)$, with $\kappa = 0.5$, i.e. $e_i = 0.5 / \sqrt{p_i}$. Check the property: under a uniform null, $E[0.5 \cdot p^{(-0.5)}] = \int_0^1 0.5 \cdot p^{(-0.5)} dp = [p^{0.5}]_0^1 = 1$. So it's a legitimate e-value. A tiny p (strong refutation) $\rightarrow$ big e; $p = 1$ (no evidence) $\rightarrow e = 0.5 < 1$ (you lose a bit of betting capital, as you should when the implication held up).

e-process / super-martingale: the *sequence* of aggregated evidence $\{E_i\}$. Popper aggregates by **multiplying**: $E_i = \prod_{s=1}^i e_s$. Under H_0 this product is a non-negative **super-martingale** ($E[E_i | \text{past}] \leq E_{i-1}$) — an *e-process*. That single property delivers everything: it stays small in expectation under the null *at every step simultaneously*, so you can peek, add experiments, and stop adaptively without breaking the guarantee.

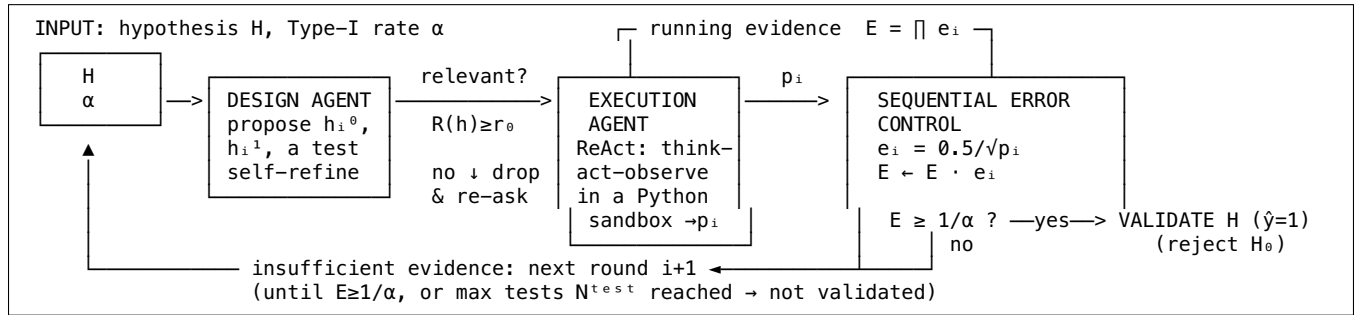
Sequential test / any-time validity / optional stopping: a test you may evaluate after every new experiment and stop the instant the evidence is conclusive, with the Type-I guarantee holding *uniformly over all stopping rules*. Classical fixed-n tests do *not* allow this; sequential e-process tests do. This is the core enabling property for an *adaptive* agent.

Type-I error: $P(\text{validate } H \mid H \text{ is actually false}) = P(\hat{y} = 1)$ under H_0 , taken over both the data randomness *and* the agent's randomness. Popper's promise: $\leq \alpha$ (default $\alpha = 0.1$).

FWER / FDR: when validating *many* hypotheses, family-wise error rate (≥ 1 false validation) is controlled by running Popper at level α/M (Bonferroni); false discovery rate (expected fraction of validated hypotheses that are false) by the e-BH procedure (Wang & Ramdas 2022), enabled for free because Popper already outputs a valid e-value per hypothesis. Both are flagged as natural extensions, not the paper's primary contribution.

The Core Algorithm

Popper is a loop of three LLM agents wrapped around a sequential test. Given a hypothesis H and a Type-I budget α , each round proposes one falsification experiment, screens it for relevance, executes it to a p-value, converts that to an e-value, multiplies it into the running evidence, and checks whether the product has crossed $1/\alpha$.



Agent	Role	How
Experiment Design Agent (A_design)	Propose the next falsification experiment: a sub-hypothesis (measurable implication), its null h_i^0 and alternative h_i^1 , and which data/test to use	LLM with domain knowledge; Self-Refine (Madaan et al.) critiques its own proposal on three axes — novelty (non-redundant with prior rounds), feasibility (answerable from available data), and logical relevance (H really implies h_i). Sees only dataset schema/metadata , never raw values — this is what keeps the e-values sequentially valid
Relevance Checker (A_rel)	Gatekeeper enforcing Assumption 1	LLM-as-judge scoring $R(h) \in [0,1]$ on a 0.1–1.0 rubric; if $R(h) < r_0$ (threshold), discard the experiment and ask the design agent for another. Prevents "falsify an irrelevant implication → inflate Type-I"
Execution Agent (A_exec)	Run the experiment, output a valid p-value p_i	ReAct (think→act→observe) inside a Jupyter/Python sandbox with pandas, statsmodels, scipy; inspects data, preprocesses, picks a test ($t / \chi^2 / \text{Mann-Whitney U} / \text{permutation} \dots$), and reports p_i in scientific notation. Up to N_{retries} on execution failure
Sequential Error Control	Aggregate evidence, decide stop/continue	Calibrate $p_i \rightarrow e_i = 0.5/\sqrt{p_i}$; multiply into $E = \prod e_i$; if $E \geq 1/\alpha$ reject H_0 (validate H); else continue

Pseudocode (Algorithm 1, condensed)

```

input: hypothesis H, dataset/world D, alpha, relevance threshold tau, max tests N_test
E <- 1                                # aggregated evidence (e-process), E_0 = 1
F_success, F_failed <- empty, empty
for i = 1 ... N_test:
    T <- A_design(H, D, F_success, F_failed)    # propose falsification experiment
    if A_rel(T) < tau:                          # relevance gate (Assumption 1)
        F_failed <- F_failed union {T}; continue
    for j = 1 ... N_retries:                    # execution with retries
        (ok, p_i) <- A_exec(T, D)
        if ok: break
    if not ok:
        F_failed <- F_failed union {T}; continue
    F_success <- F_success union {T}
    e_i <- 0.5 / sqrt(p_i)                      # p-to-e calibrator, kappa = 0.5
    E <- E * e_i                                # multiply (super-martingale)
    if E >= 1/alpha:                            # sufficient evidence
        return VALIDATED (y-hat = 1)           # i.e. reject H_0
return NOT VALIDATED (y-hat = 0)              # budget exhausted
  
```

Why this controls Type-I error (Theorem 4, in plain steps)

The guarantee $P(\hat{y} = 1) \leq \alpha$ under H_0 rests on three assumptions and a one-line martingale argument:

- **Assumption 1 (Implication):** H_0 true $\Rightarrow$ every h_i^0 true. (Enforced by the relevance checker.)
- **Assumption 2 (Sequential validity):** each e_i satisfies $E[e_i \mid \text{data so far}] \leq 1$ under h_i^0 . Popper engineers this by giving the design agent only *schema/metadata* of unused datasets — the choice of the next test cannot peek at the values it will be tested on, so the p-value stays conditionally valid. (For a static database this is automatic; for actively-collected data it holds because the data is gathered *after* the design step.)
- **Assumption 3 (Optional stopping):** the stopping decision depends only on data seen so far (the running E is measurable) — a valid stopping time.

Then the proof: under H_0 , $E[E_i \mid F_{i-1}] = E_{i-1} \cdot E[e_i \mid F_{i-1}] \leq E_{i-1}$, so $\{E_i\}$ is a **non-negative super-martingale** with $E_0 = 1$. By **Doob's optional-stopping theorem**, at *any* stopping time τ , $E[E_\tau] \leq 1$. By **Markov's inequality**, $P(\hat{y}=1) = P(E_\tau \geq 1/\alpha) \leq \alpha \cdot E[E_\tau] \leq \alpha$. Done. The agent may run any number of experiments and stop whenever — the cap holds because multiplying e-values can only inflate your "wealth" by chance with probability $\leq \alpha$ when the null is true.

Numerical walkthrough of the sequential error control

Concrete run on $H = \text{"Gene ZAP70 regulates the production of Interleukin-2,"}$ $\alpha = 0.1$, so the rejection threshold is $1/\alpha = 10$. Calibrator: $e = 0.5/\sqrt{p}$. Aggregation: $E = \prod e$. (These are the actual cumulative e-values from the paper's Table 1.)

Round	Falsification experiment (measurable implication)	p-value	$e=0.5/\sqrt{p}$	$E=\prod e$	
1	ZAP70 protein-protein interactions with IL-2 pathway components (mass-spec) – no enrichment found	1.0	0.50	0.50	
2	ZAP70 expression correlates with IL-2-pathway genes across tissues (eQTL/expression)	8.8e-3	5.33	2.67	
3	Genetic variants affecting ZAP70 also change IL-2-pathway activity (eQTL) – inconclusive	~1.0	0.50	1.33	← evidence DROPS back
4	Rare missense variants in ZAP70 associate with immune phenotypes (GeneBASS) – strong association	4.7e-4	23.06	30.78	← $E \geq 10$ VALIDATE
Decision: at round 4, $E = 30.78 \geq 10 = 1/\alpha \rightarrow$ reject H_0 , hypothesis VALIDATED.					

What to notice, because it's the whole point of using e-values instead of "take the smallest p-value":

- **Round 1's failed refutation costs you.** The implication held up ($p=1.0$), so $e=0.5 < 1$: your evidence *shrinks* to 0.5. A hypothesis that survives a sincere refutation attempt has *not* gained support — Popperian honesty, encoded numerically.
- **Round 3 is inconclusive ($p \approx 1$) and drags E from 2.67 down to 1.33.** Evidence is allowed to go *backwards*. There is no cherry-picking: you can't quietly discard the rounds that hurt you, because the product includes them all.
- **No multiplicity correction is bolted on.** With a "min p-value" rule, four tries at $\alpha=0.1$ would give a false-positive rate far above 10%. Here, four tries are *already* honest: had H_0 been true, the four e-values would average ≤ 1 each and the product would clear 10 with probability ≤ 0.1 , by construction. The agent ran an *adaptive* number of experiments (it would have stopped at any round where E hit 10) and the 10% cap still holds.
- **The threshold is interpretable.** " $E \geq 1/\alpha$ " means "you turned \$1 into \$10 betting against the null" — a 10× return is hard to fake when the null is true.

Results Across Six Domains

Popper is evaluated on two benchmark collections — and the cross-domain breadth is the reason a non-biology reader should care.

- **TargetVal** (deep biology): aggregates **22 tables totaling ~85 million records** from GTEx, the GWAS Catalog (MacArthur 2017), and BioGrid (Oughtred 2019). Hypotheses follow "Gene A regulates Phenotype B"; two sub-tasks **TargetVal-IL2** and **TargetVal-IFNG**; ground truth from genome-wide CRISPR screens (Schmidt 2022). 20 positive hypotheses, 50 constructed negatives (via column permutation).
- **DiscoveryBench** (Majumder 2024): spans **six domains — sociology, biology, humanities, economics, engineering, and meta-science** — yielding **86 non-null hypotheses** grounded in peer-reviewed research. This is where the domain-generality claim lives: same falsification machinery, no biology-specific tooling.

Headline numbers ($\alpha = 0.1$; mean $\pm$ sd over 5 runs; default LLM Claude-Sonnet-3.5):

Method	Type-I error (≤ 0.1 = controlled)			Power (higher = better)		
	DiscBench	TargetVal-IL2	TargetVal-IFNG	DiscBench	IL2	IFNG
ReAct	0.078	0.000	0.000	0.383	0.010	0.020
Self-Refine	0.117	0.100	0.067	0.476	0.183	0.067
CodeGen (o1)	0.248x	0.013	0.000	—	0.250	0.183
Fisher combined	0.311x	0.264x	0.173x	—	—	—
LLM-Likelihood ratio	0.152x	0.016	0.180x	0.428	0.185	—
POPPER (ours)	0.103	0.082	0.085	0.638	0.580	0.591

- **Type-I error is controlled across all three settings** (0.103, 0.082, 0.085 — all within 1 sd of the 0.1 nominal), while most baselines either blow past it (Fisher's combined test 0.26–0.31; CodeGen-o1 0.25) or only "control" it by never rejecting anything (ReAct: Type-I 0.000 *because* power is ~ 0.01 — useless).
- **Highest power among methods that actually control Type-I.** vs ReAct on DiscoveryBench: **+66.5% power** (0.638 / 0.383). vs Self-Refine on TargetVal-IL2: **3.17× power** (0.580 / 0.183).
- **Ablations confirm each piece earns its place.** Removing the relevance checker (**Popper-NoReleCheck**) inflates Type-I (e.g. 0.340 on IL2, 0.300 on IFNG) — irrelevant tests break Assumption 1. Swapping e-values for Fisher's combined test loses control entirely. Swapping ReAct for plain code-gen (**Popper-CodeGen**) loses power. The whole stack is load-bearing.
- **Sensitivity:** sweeping α from 0.10→0.20 tracks the empirical Type-I error along the diagonal (calibration is honest, not coincidental); raising the max-test budget *raises power while keeping Type-I flat* — more compute → more discoveries, safely.

vs human experts (9 PhD-level biostatisticians / computational biologists), on TargetVal-IL2:

	Type-I error	Power	Time	Lines of code	# stat tests
Human experts	22.2%	66.7%	1x	1x	1x
POPPER	11.1%	66.7%	9.7× FASTER	3.6× more	2.5× more

Statistically indistinguishable accuracy (small n), **$\sim 10\times$ faster** (the abstract's "reducing time by 10 folds"), with *better* Type-I control and more thorough testing. Qualitatively, Popper and humans converged on the same families of tests (permutation, Fisher's, eQTL, χ^2) and the same falsification experiment types (expression enrichment, interaction networks, variant overlap).

LLM backbone matters a lot. o1 gave the best Type-I control (0.091 DiscoveryBench / 0.031 IL2) and best DiscoveryBench power (0.654); Sonnet-3.5 led IL2 power (0.580). Weak models break the guarantee:

Claude-Haiku-3.5 had Type-I 0.230 — it cannot reliably produce valid p-values or honor the implication structure. Rigorous reasoning + coding is a prerequisite, not a nicety.

Where Popper still fails (error analysis over 128 failed logs, auto-labeled by o1, human-verified): misinterpreting the context/validity of a p-value **35.9%**, ineffective test selection **28.1%**, falsification test that *breaks the implication* (Assumption-1 violation) **17.2%**, incorrect test implementation **8.6%**, failure to locate relevant data **7.0%**. **Hallucination was 0.8% (1 case), and no p-hacking was observed** — the structure successfully suppresses the failure mode you'd most fear from an LLM doing statistics.

Limitations (stated by the authors)

- **Type-I control $\neq$ true discovery.** Rejecting H_0 caps the false-validation *rate*, but a validated hypothesis is not guaranteed true — in the extreme where every input hypothesis is null, *every* validation Popper emits is false, just at rate $\leq \alpha$ each. Treat "validated" as "survived rigorous falsification," not "proven." (The authors note FDR/FWER extensions via Bonferroni α/M or e-BH for the many-hypotheses regime.)

- **Reasoning about p-value *validity* is the dominant error source (35.9%).** The agent can run a technically-correct test whose p-value doesn't mean what it thinks — e.g. computing an eQTL p-value for "RAB39A expression in neutrophils" and treating it as evidence that "RAB39A regulates IL-2," when the implication doesn't actually hold. Effect-size/p-value semantics remain hard.
- **The relevance checker is good but not perfect.** It agrees with human raters reasonably (Spearman $\rho = 0.55$, $p = 5e-6$; it rated 84% of tests "Strongly Relevant" vs. humans' 77%) but slightly *over-estimates* relevance, which is exactly the direction that risks Assumption-1 violations.
- **Needs a strong backbone.** Below a capability threshold (Haiku-3.5) Type-I control simply fails. The guarantee is conditional on the agent producing genuinely valid p-values.
- **Current instantiation queries static databases.** The framework is defined for live data collection and simulation too, but the demonstrated runs draw from fixed corpora; real-world experimental execution (robots, wet-lab) is described in principle, not shown.

Popper vs. Alternatives

The cleanest way to place Popper: it is the **validation** complement to the **generation** systems. They emit hypotheses; Popper decides — with a Type-I guarantee — whether to believe them.

System	Primary job	Free-form hyp. input?	Statistical error control?	Adaptive / optional stop?	Domains
POPPER	VALIDATION (falsify+test)	✓	✓ (e-process, Type-I $\leq \alpha$)	✓ (any-time valid)	biology, econ, sociology, +3
Robin	generation + wet-lab loop	x (it generates)	x (LLM-judge + BTL rank)	~ (refine on results)	wet-lab biology
AI-Scientist	generation + full automation	x	x	~	computational ML
AI Co-Scientist	generation / debate	x	x (debate / evolve)	x	biomedical
DiscoveryBench (as a system)	benchmark / data-driven	—	partial (task metric)	x	6 (it supplies Popper's hyps)

- **Robin / AI-Scientist / AI Co-Scientist rank candidates with an LLM judge** (Robin even fits a Bradley–Terry–Luce model over pairwise preferences — the same BTL you know from RLHF reward modeling and DPO's $P(\text{prefer } A) = \sigma(r_A - r_B)$). That gives a *relative ordering*, not a *calibrated false-positive rate*. Popper's e-process gives the thing BTL ranking cannot: an absolute, frequentist guarantee that you validate a false hypothesis $\leq \alpha$ of the time. Different statistical object — a betting-wealth e-value, not a preference logit.
- **Natural pipeline:** Robin/AI Co-Scientist *propose and rank* → Popper *adjudicates the top candidates* before anyone spends wet-lab money or makes a policy decision. Robin closes the loop in biology; Popper makes the loop's "is this real?" step honest, and does it in economics and sociology too.
- **When Popper is the wrong tool:** if the hypothesis has *no* testable implication reducible to a data query / statistical test, or if you need novel mechanism *generation* (Popper validates, it does not invent), or if your backbone LLM is weak (the guarantee evaporates).

Practical Considerations

- **The reusable idea is the p-to-e-to-product pipeline.** Any time an agent runs *multiple adaptive tests and decides when to stop*, classical p-values give you no honest aggregate. Convert each valid p-value with a calibrator ($e = 0.5/\sqrt{p}$ is a fine default), multiply, and reject at $1/\alpha$. You get multiplicity *and* optional-stopping protection for free, with no Bonferroni bookkeeping. This transfers to A/B testing, monitoring, and any LLM-judge-style eval where you'd otherwise "keep testing until significant."

- **Guard the implication, not just the test.** The single most important non-obvious component is the relevance checker. Without it (ablation), Type-I error roughly *triples*. If you build something like this, budget real effort on "does H actually imply this sub-test?" — a valid statistical test of an *irrelevant* implication silently destroys the guarantee.
- **Keep the proposer blind to the values it will test on.** Sequential validity (Assumption 2) hinges on the design agent seeing only schema/metadata of unused data. Letting it peek at raw values to choose the next test is a subtle way to invalidate every downstream e-value.
- **Tune two knobs.** α (default 0.1) trades discovery rate against false validations; the max-test budget trades compute against power (more tests $\rightarrow$ more power, Type-I unaffected). Both behave monotonically and predictably.
- **Use a capable backbone and expect p-value-semantics errors.** Pick an o1-class or Sonnet-class model; below that the guarantee fails. Even with a good model, the dominant residual failure is misreading what a p-value means, so a human spot-check on validated hypotheses is worth the cost.
- **Don't over-read a "validated" verdict.** It means "survived sequential falsification at level α ," which is a real, rare, and useful signal — but for acting on many validated hypotheses at once, wrap Popper in FDR control (e-BH) rather than trusting each verdict independently.

Key Papers

1. Huang, Jin, R. Li, M. Y. Li, Candès, Leskovec. *Automated Hypothesis Validation with Agentic Sequential Falsifications*. arXiv 2502.09858 (2025). <https://arxiv.org/abs/2502.09858>
2. Vovk & Wang. *E-values: Calibration, combination and applications*. Annals of Statistics 49(3):1736–1754 (2021). — the e-value definition and the p-to-e calibrator Popper uses.
3. Grünwald, de Heide & Koolen. *Safe testing*. IEEE ITA (2020), arXiv 1906.07801. — any-time-valid testing as betting; the e-process / optional-stopping foundation.
4. Shafer. *The language of betting as a strategy for statistical and scientific communication*. arXiv 1903.06991 (2019). — betting interpretation of e-values.
5. Wang & Ramdas. *False discovery rate control with e-values*. JRSS-B 84(3):822–852 (2022). — e-BH; the route from Popper's per-hypothesis e-values to FDR control.
6. Popper. *The Logic of Scientific Discovery*. Hutchinson (1959) / Routledge (2005). — the falsification philosophy the system implements.
7. Madaan et al. *Self-Refine: Iterative Refinement with Self-Feedback*. NeurIPS 36 (2024). — the design agent's self-critique loop.
8. Yao et al. *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR 2023, arXiv 2210.03629. — the execution agent's think-act-observe loop.
9. Majumder et al. *DiscoveryBench: Towards Data-Driven Discovery with Large Language Models*. arXiv 2407.01725 (2024). — the six-domain hypothesis benchmark.
10. Lu et al. *The AI Scientist*. arXiv 2408.06292 (2024); Gottweis et al. *Towards an AI Co-Scientist*. arXiv 2502.18864 (2025); Ghareeb et al. *Robin*. arXiv 2505.13400 (2025). — the generation-side systems Popper complements.